임베디드 시스템

Embedded Systems

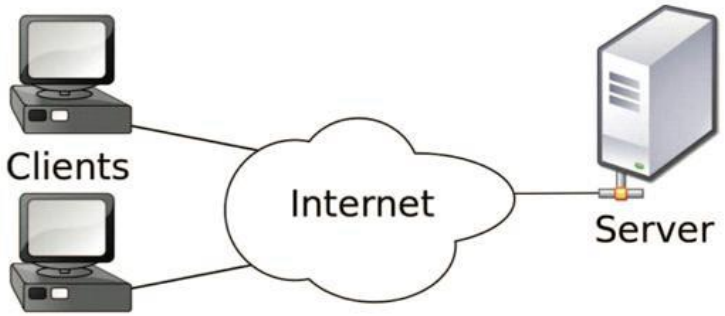
Web Server

강원대학교 컴퓨터공학과 권용훈

라즈베리 파이 웹 서버 환경구축(1/8)

웹 서버 환경 구축

- APM 설치
  - APM – Apache + php + MySQL
  - Apache: 서버용 소프트웨어
  - php: 웹 페이지 구축을 위한 프로그래밍 언어
  - MySQL: 데이터 베이스 언어




```
pi@raspberrypi ~/pi $
pi@raspberrypi ~/pi $ sudo apt-get install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
apache2 is already the newest version.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
pi@raspberrypi ~/pi $
```

라즈베리 파이 웹 서버 환경구축(2/8)

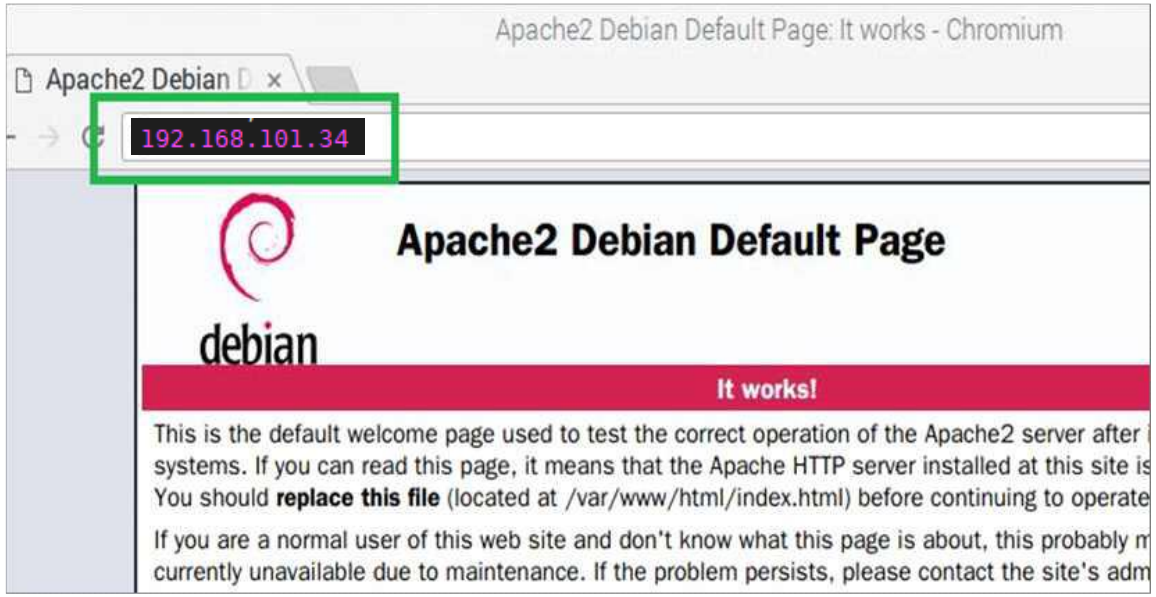
Apache 설치 확인

- ifconfig: 내부 IP 확인

```
pi@raspberrypi:~ $ ifconfig
eth0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        ether b8:27:eb:2e:92:3c  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 61  bytes 9836 (9.6 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 61  bytes 9836 (9.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.101.34  netmask 255.255.255.0  broadcast 192.168.101.255
        inet6 fe80::e953:6308:587b:c4bb  prefixlen 64  scopeid 0x20<link>
        ether b8:27:eb:7b:c7:69  txqueuelen 1000  (Ethernet)
        RX packets 296190  bytes 427686547 (407.8 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 162964  bytes 17099025 (16.3 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

브라우저 실행

- IP주소 입력 후 페이지 확인



라즈베리 파이 웹 서버 환경구축(3/8)

Document Root 생성하기

- 본인의 디렉토리에 ./www 폴더 생성

```
mkdir ./www
sudo chmod 777 ./www
```

- apache2.conf 설정파일 변경

```
sudo nano /etc/apache2/apache2.conf
```

```
# Sets the default security model of the Apache2 HTTPD server. It does
# not allow access to the root filesystem outside of /usr/share and /var/www.
# The former is used by web applications packaged in Debian,
# the latter may be used for local directories served by the web server. If
# your system is serving content from a sub-directory in /srv you must allow
# access here, or in any related virtual host.
<Directory />
        Options FollowSymLinks
        AllowOverride None
        Require all denied
</Directory>

<Directory /usr/share>
        AllowOverride None
        Require all granted
</Directory>

<Directory /home/pi/www>
        Options Indexes FollowSymLinks
        AllowOverride None
        Require all granted
</Directory>
```

본인의 디렉토리 경로로 작성

- 000-default.conf 설정파일 변경

```
sudo nano /etc/apache2/sites-available/
000-default.conf
```

```
<VirtualHost *:80>
        # The ServerName directive sets the request scheme, hostname and port that
        # the server uses to identify itself. This is used when creating
        # redirection URLs. In the context of virtual hosts, the ServerName
        # specifies what hostname must appear in the request's Host: header to
        # match this virtual host. For the default virtual host (this file) this
        # value is not decisive as it is used as a last resort host regardless.
        # However, you must set it for any further virtual host explicitly.
        #ServerName www.example.com

        ServerAdmin webmaster@localhost
        #DocumentRoot /var/www/html
        DocumentRoot /home/pi/www

        # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
        # error, crit, alert, emerg.
        # It is also possible to configure the loglevel for particular
        # modules, e.g.
        #LogLevel info ssl:warn

        ErrorLog ${APACHE_LOG_DIR}/error.log
        CustomLog ${APACHE_LOG_DIR}/access.log combined

        # For most configuration files from conf-available/, which are
        # enabled or disabled at a global level, it is possible to
        # include a line for only one particular virtual host. For example the
        # following line enables the CGI configuration for this host only
        # after it has been globally disabled with "a2disconf".
        #Include conf-available/serve-cgi-bin.conf
</VirtualHost>

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet
```

라즈베리 파이 웹 서버 환경구축(4/8)

Document Root 생성하기

- default-ssl.conf 설정파일 변경

```
sudo nano /etc/apache2/sites-available/
default-ssl.conf
```

```
<IfModule mod_ssl.c>
        <VirtualHost _default_:443>
                ServerAdmin webmaster@localhost

                #DocumentRoot /var/www/html
                DocumentRoot /home/pi/www

                # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
                # error, crit, alert, emerg.
                # It is also possible to configure the loglevel for particular
                # modules, e.g.
                #LogLevel info ssl:warn

                ErrorLog ${APACHE_LOG_DIR}/error.log
                CustomLog ${APACHE_LOG_DIR}/access.log combined

                # For most configuration files from conf-available/, which are
                # enabled or disabled at a global level, it is possible to
                # include a line for only one particular virtual host. For example the
                # following line enables the CGI configuration for this host only
                # after it has been globally disabled with "a2disconf".
                #Include conf-available/serve-cgi-bin.conf

                #   SSL Engine Switch:
                #   Enable/Disable SSL for this virtual host.
                SSLEngine on
```

- 변경 사항 적용하기

```
sudo /etc/init.d/apache2 reload
sudo /etc/init.d/apache2 restart
sudo service apache2 restart
```

- www 디렉토리에 index.html 파일 추가

라즈베리 파이 웹 서버 환경구축(5/8)

php

- php 패키지 설치

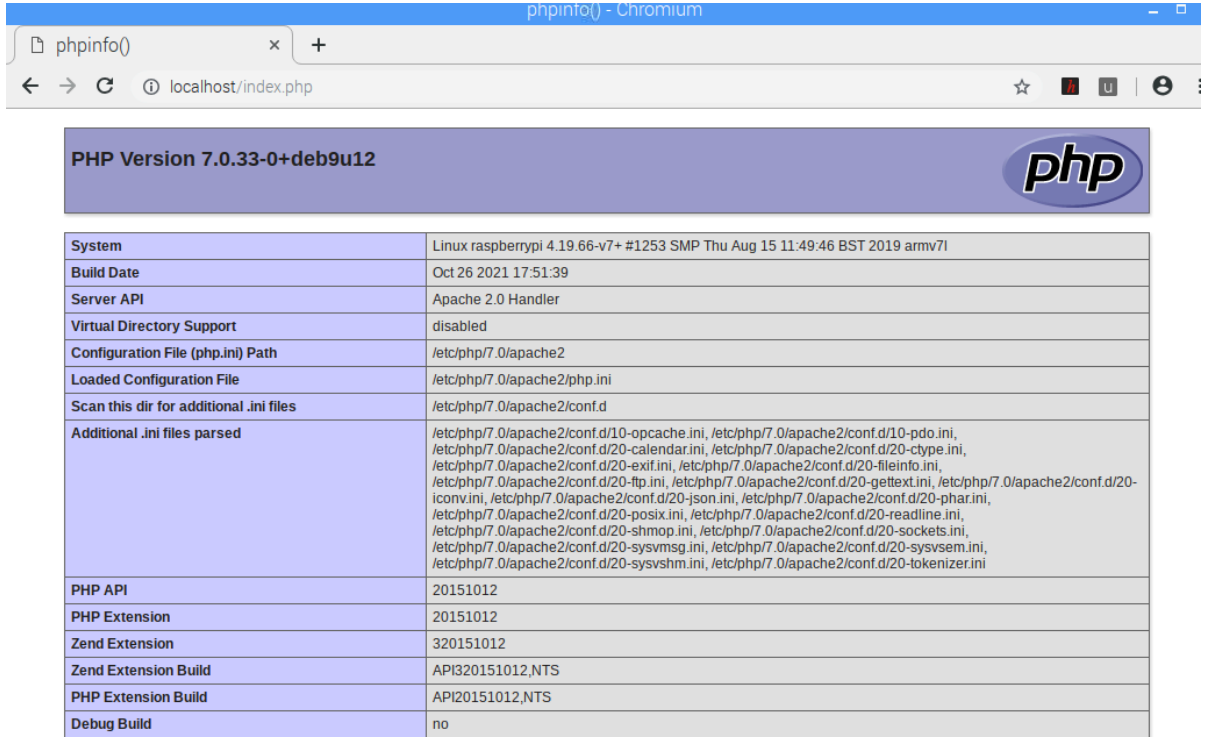
```
sudo apt-get install php7.0 php7.0-common
libapache2-mod-php7.0
```

- php 설치 확인

```
sudo nano ./www/index.php
```

```
<?php
        phpinfo();
?>
```

- 웹 브라우저를 통하여 확인



System	Linux raspberrypi 4.19.66-v7+ #1253 SMP Thu Aug 15 11:49:46 BST 2019 armv7l
Build Date	Oct 26 2021 17:51:39
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/7.0/apache2
Loaded Configuration File	/etc/php/7.0/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/7.0/apache2/conf.d
Additional .ini files parsed	/etc/php/7.0/apache2/conf.d/10-opcache.ini, /etc/php/7.0/apache2/conf.d/10-pdo.ini, /etc/php/7.0/apache2/conf.d/20-calendar.ini, /etc/php/7.0/apache2/conf.d/20-ctype.ini, /etc/php/7.0/apache2/conf.d/20-exif.ini, /etc/php/7.0/apache2/conf.d/20-fileinfo.ini, /etc/php/7.0/apache2/conf.d/20-ftp.ini, /etc/php/7.0/apache2/conf.d/20-gettext.ini, /etc/php/7.0/apache2/conf.d/20-iconv.ini, /etc/php/7.0/apache2/conf.d/20-json.ini, /etc/php/7.0/apache2/conf.d/20-phar.ini, /etc/php/7.0/apache2/conf.d/20-posix.ini, /etc/php/7.0/apache2/conf.d/20-readline.ini, /etc/php/7.0/apache2/conf.d/20-shmop.ini, /etc/php/7.0/apache2/conf.d/20-sockets.ini, /etc/php/7.0/apache2/conf.d/20-sysvmsg.ini, /etc/php/7.0/apache2/conf.d/20-sysvsem.ini, /etc/php/7.0/apache2/conf.d/20-sysvshm.ini, /etc/php/7.0/apache2/conf.d/20-tokenizer.ini
PHP API	20151012
PHP Extension	20151012
Zend Extension	320151012
Zend Extension Build	API320151012,NTS
PHP Extension Build	API20151012,NTS
Debug Build	no

라즈베리 파이 웹 서버 환경구축(6/8)

MySQL(MariaDB)

- 패키지 설치

sudo apt-get install **mariadb-server mariadb-client**

sudo apt-get install **libmariadbclient-dev**

sudo apt-get install **php7.0-mysql**

mysql --version

- 서버 접속

sudo mysql –uroot

MariaDB [(mysql)] > **use mysql**

MariaDB [(mysql)] > **select user, host, password from user;**

```
MariaDB [(none)]> use mysql
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [mysql]> select user, host, password from user;
+------+-----------+----------+
| user | host      | password |
+------+-----------+----------+
| root | localhost |          |
+------+-----------+----------+
```

라즈베리 파이 웹 서버 환경구축(7/8)

MySQL(MariaDB)

- 패스워드 설정

```
MariaDB [(mysql)] > update user set
password=PASSWORD('1234')
where user='root';

MariaDB [(mysql)] > flush privileges;
```

```
MariaDB [mysql]> update user set password=PASSWORD('1234') where user='root';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

MariaDB [mysql]> flush privileges;
Query OK, 0 rows affected (0.01 sec)
```

- 종료

```
MariaDB [(mysql)] > quit
```

- 외부 접근 허용 설정

```
cd /etc/mysql/mariadb.conf.d

ls

sudo nano 50-server.cnf
```

```
#
# * Basic Settings
#
user            = mysql
pid-file        = /var/run/mysqld/mysqld.pid
socket          = /var/run/mysqld/mysqld.sock
port            = 3306
basedir         = /usr
datadir         = /var/lib/mysql
tmpdir          = /tmp
lc-messages-dir = /usr/share/mysql
skip-external-locking

# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
#bind-address            = 127.0.0.1
```

라즈베리 파이 웹 서버 환경구축(8/8)

MySQL(MariaDB)

- 설정 변경 후 재접속

```
sudo service mysql restart

sudo mysql –u root –p
```

- 외부 권한 부여

```
MariaDB [(none)] > show grants;

MariaDB [(none)] > grant all privileges on *.* to 'root'@'%' identified by '1234';

MariaDB [(none)] > flush privileges;
```

- 접근 사용자 정보 확인

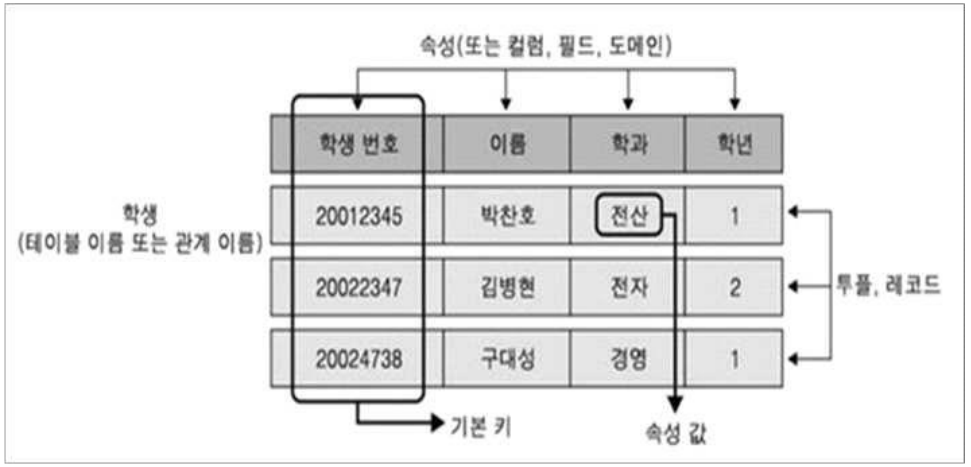
```
MariaDB [(none)] > use mysql;

MariaDB [(mysql)] > select user, host, password from user;
```

```
MariaDB [mysql]> select user, host, password from user;
+------+-----------+-------------------------------------------+
| user | host      | password                                  |
+------+-----------+-------------------------------------------+
| root | localhost | *A4B6157319038724E3560894F7F932C8886EBFCF |
| root | %         | *81F5E21E35407D884A6CD4A731AEBFB6AF209E1B |
+------+-----------+-------------------------------------------+
2 rows in set (0.00 sec)
```

데이터 베이스 생성(1/2)

MySQL을 사용한 DB 생성

- MySQL은 관계형 데이터 베이스의 일종이다.
- 테이블 형태로 데이터를 저장하고, 쉽고 간단하게 조작하여 원하는 데이터를 검색, 정렬할 수 있다.




```
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
3 rows in set (0.00 sec)

MariaDB [(none)]>
```

데이터 베이스 생성(2/2)

MySQL을 사용한 DB 생성

- 새로운 DB 생성하기

```
MariaDB [(none)] > create database 생성할_DB_이름;
```

이름은 본인의 'EMB+학번' 으로 할 것

- 새로운 테이블 생성

```
MariaDB [(none)] > use 생성한_DB_이름 ;

MariaDB [(EMB401581)] > create table ...
```

```
MariaDB [(EMB401581)] > create table sensor (
time datetime primary key,
ultrasonic int(3) not null,
illum int(3) not null);
```

- 생성된 테이블 확인

```
MariaDB [EMB401581]> show tables;
+---------------------+
| Tables_in_EMB401581 |
+---------------------+
| sensor              |
+---------------------+
1 row in set (0.00 sec)

MariaDB [EMB401581]> desc sensor;
+------------+----------+------+-----+---------+-------+
| Field      | Type     | Null | Key | Default | Extra |
+------------+----------+------+-----+---------+-------+
| time       | datetime | NO   | PRI | NULL    |       |
| ultrasonic | int(3)   | NO   |     | NULL    |       |
| illum      | int(3)   | NO   |     | NULL    |       |
+------------+----------+------+-----+---------+-------+
3 rows in set (0.01 sec)
```

데이터 베이스 API(1/4)

MySQL C 라이브러리

- MySQL 자료형

타 입	기 능
MYSQL	MySQL DB와 연결을 관리하는 구조체
MYSQL_RES	쿼리 결과를 나타내는 자료형
MYSQL_ROW	MYSQL_RES에서 하나의 레코드씩 값을 저장하는 자료형
MYSQL_FILED	필드에 대한 정보를 저장하는 구조체

*** Query문 실행**

```
int mysql_query (MYSQL *mysql, const char *stmt_str)
```

- 매개변수 : stmt_str ← ;를 포함하지 않은 쿼리 문
- 반환 : 성공 - 0

*** 바이너리 데이터 포함한 Query문 실행**

```
int mysql_real_query (MYSQL *mysql,
                      const char *stmt_str,
                      unsigned long length)
```

- 반환 : 성공 - 0

*** MYSQL 객체 할당 및 초기화**

```
MYSQL  *mysql_init(MYSQL *mysql)
```

- 반환 : MYSQL Handle

*** MYSQL 데이터베이스 엔진 연결 설정 및 시도**

```
MYSQL *mysql_real_connect(MYSQL *mysql,
                          const char *host,
                          const char *user, const char *passwd,
                          const char *db,
                          unsigned int port,
                          const char *unix_socket,
                          unsigned long client_flag)
```

- 반환 : 성공 - MYSQL Handle / 실패 - NULL

*** MYSQL 객체 해제**

```
void mysql_close(MYSQL *mysql)
```

- 반환 : 없음

데이터 베이스 API(2/4)

MySQL C 라이브러리

* **Query 결과값을 모두 가져오는 함수**
  MYSQL_RES *mysql_store_result (MYSQL *mysql)
  - 반환 : 성공 – MYSQL_RES / 실패 - NULL

* **Query 결과값을 1개만 가져오는 함수**
  MYSQL_RES *mysql_use_result (MYSQL *mysql)
  - 반환 : 성공 – MYSQL_RES / 실패 – NULL

* **결과값이 저장된 메모리 할당 해제 함수**
  void mysql_free_result (MYSQL *mysql)
  - 반환 : 없음ㅍ

* **Query 결과 데이터 개수**
  rows mysql_num_rows (MYSQL_RES *result)

* **Query의 필드 개수**
  unsigned int mysql_num_fields (MYSQL_RES *result)

* **Query의 필드 이름 출력**
  void mysql_free_result (MYSQL_RES *result , i)
  - 반환 : 없음

•**Query 결과를 배열에 담기**

array mysql_fetch_array (result , result_type = MYSQL_BOTH)
- result : mysql_query() 결과값
- result_type : 배열로 저장 시 기준 지정
  MYSQL_ASSOC : 컬럼 명
  MYSQL_NUM : 컬럼 인덱스
  MYSQL_BOTH : 두 가지 모두 저장
- 반환 : 성공 – MYSQL_RES / 실패 - NULL

데이터 베이스 API(3/4)

```c
#include <wiringPi.h>
#include <stdio.h>
#include <stdlib.h>
#include <mysql/mysql.h>
#include <time.h>
#include <math.h>
//- raspiDB 데이터베이스 관련 상수 ------------------------
#define DB_HOST "000" //- RaspberryPi IP 입력
#define DB_USER "root"
#define DB_PASS "root"
#define DB_NAME "-----"
//- 라이브러리 포함 ------------------------------------
---
MYSQL *connector;
MYSQL_RES *result;
MYSQL_ROW row;
//- 데이터 저장 구조체 선언 ------------------------------
typedef struct _sensor {
    int ultrasonic;
    int illum;
}SENSOR;
//- 전역변수 선언 --------------------------------------
---
int gUltra=50, gIll = 100;
int main(void){
    printf( "Raspberry Pi - MySQL DB Test \n" );
    if ( wiringPiSetup() == -1 ) exit( 1 );
    //- 연결
```

```c
connector = mysql_init(NULL);
    if(!mysql_real_connect(connector, DB_HOST, DB_USER,
DB_PASS,DB_NAME, 3306, NULL, 0)){
        fprintf( stderr, "%s \n", mysql_error(connector));
        return 0;
    }else{
        printf("MySQL(raspiDB) opened! \n");
    }
    while(1){
        char query[1024];
        SENSOR gData = {gUltra++,gIll++ };
        sprintf(query, "insert into sensor values (now(), %d, %d, %d)",
gData.ultrasonic, gData.illum, 0);
        if(mysql_query(connector, query)){
            fprintf(stderr, "%s \n", mysql_error(connector));
            printf("Write DB error. \n");
        }else{
            printf("Insert OK!! \n");
        }
    delay(1000);
    }
    mysql_close(connector);
    return(0);
}
```

gcc **–lmariadbclient** –lwiringPi –o EX_DB EX_MyDB.c

데이터 베이스 API(4/4)

결과 확인

실행 전

```
Database changed
MariaDB [EMB401581]> select * from sensor;
Empty set (0.00 sec)

MariaDB [EMB401581]>
```

실행 후

```
pi@raspberrypi:~ $ sudo ./EX_DB
Raspberry Pi - MySQL DB Test
MySQL(raspiDB) opened!
Insert OK!!
Insert OK!!
Insert OK!!
Insert OK!!
Insert OK!!
Insert OK!!
Insert OK!!
```

```
Database changed
MariaDB [EMB401581]> select * from sensor;
+---------------------+------------+-------+
| time                | ultrasonic | illum |
+---------------------+------------+-------+
| 2023-10-12 16:12:26 |         50 |   100 |
| 2023-10-12 16:12:27 |         51 |   101 |
| 2023-10-12 16:12:28 |         52 |   102 |
| 2023-10-12 16:12:29 |         53 |   103 |
| 2023-10-12 16:12:30 |         54 |   104 |
| 2023-10-12 16:12:31 |         55 |   105 |
| 2023-10-12 16:12:32 |         56 |   106 |
+---------------------+------------+-------+
7 rows in set (0.01 sec)
```

웹 서버 연동(1/2)

php 기본 문법

```
<?php
    CODE
?>
```

```
<?php
    echo( " " ) ;          <- ; 명령어 끝 세미콜론
?>
```

웹 페이지나 콘솔에 표시

```
$temp = 10.5;
echo "$temp 도 입니다. <br>";
echo '$temp 도 입니다. <br>';
```

```
<?php
    echo "Tmp = " . $Tmp . "<br>" ;
?>
```

문자열 이어주기 기능

```
$temp = 10.5;
echo "$temp 도 입니다. " . $data . "<br>";
```

```
<?php
    print ( " 출력문 " );
    print "출력문";
?>
```

웹 서버 연동(1/2)

php와 MySQL 연동

* PHP와 MYSQL 연결 함수
  conn_object **mysqli_connect**(db_host, user_id, user_pw, db_name, port)
  - 성공 : conn_object / 실패 : FALSE

* PHP와 MYSQL 연결 해제 함수
  mysqli_close(conn_object)

* PHP와 MYSQL 연결 에러 처리 함수
  int **mysqli_connect_errno**(void) : 0 – 연결 성공
  string **mysqli_connect_error**(void)

* PHP와 MYSQL 데이터 조회 함수
  conn_object **mysqli_query**(conn_object , string query)
  - 성공 : mysql_result object / 실패 : FALSE

* PHP와 MYSQL 데이터 조회 개수 함수
  int **mysqli_num_rows**(mysql_result_object)

* PHP와 MYSQL 데이터 결과 함수
  array **mysqli_fetch_array**(mysql_result_object)
  - 성공 : array

  array **mysqli_fetch_row**(mysql_result_object)
  - 성공 : array

  array **mysqli_fetch_assoc**(mysql_result_object)
  - 성공 : array

웹 서버 연동(2/2)

```
<html>
<title> [ PHP & Raspberry Pi ] </title>
<body>
<script language='javascript'>
   window.setTimeout('window.location.reload()', 1000);
</script>
<h3> PHP & MySQL View </h3>
<?php
//- raspiDB 관련 변수 선언 ----------------------------------------------------
$db_host = "주소";
$db_user = "root";
$db_passwd = "패스워드";
$db_name = " ";
$db_port = 3306 ;
//- raspiDB 연결 ----------------------------------------------------------------
echo "(1) DB Connecting ----------------------<br>";
$conn = mysqli_connect( $db_host, $db_user, $db_passwd, $db_name,
$db_port );
if(!$conn) {
    echo " ₩tDB Connect : Fail ~!! [ " . mysqli_connect_errno() . " ] ₩t";
    echo mysqli_connect_error() . "<br><br>";
}else {
    echo " ₩tDB Connected. OK~!!<br><br>";
}
//- raspiDB 데이터 조회----------------------------------------
echo "(2) DB Query --------------------------------<br>";
$query = "select * from sensor";
$result = mysqli_query($conn, $query);
$rows = mysqli_num_rows($result);
echo "Rows = " . $rows."<br><br>";
//- Disconnect ----------------------------------------------------
echo "(3) DB Disconnect ----------------------<br>";
    mysqli_free_result($result);
    mysqli_close($connect);
    echo " ₩tDB Disconnect OK ~ !!<br>";
?>
</body>
</html>
```

Not secure | 192.168.101.34/viewDB.php

PHP & MySQL View

(1) DB Connecting ----------
DB Connected. OK~!!

(2) DB Query ----------
Rows = 0

(3) DB Disconnect ----------
DB Disconnect OK ~ !!

고생하셨습니다

Thank you!